



# PyTorch

**PyTorch** is a machine learning library based on the Torch library,<sup>[4][5][6]</sup> used for applications such as computer vision and natural language processing,<sup>[7]</sup> originally developed by Meta AI and now part of the Linux Foundation umbrella.<sup>[8][9][10][11]</sup> It is one of the most popular deep learning frameworks, alongside others such as TensorFlow,<sup>[12]</sup> offering free and open-source software released under the modified BSD license. Although the Python interface is more polished and the primary focus of development, PyTorch also has a C++ interface.<sup>[13]</sup>

A number of pieces of deep learning software are built on top of PyTorch, including Tesla Autopilot,<sup>[14]</sup> Uber's Pyro,<sup>[15]</sup> Hugging Face's Transformers,<sup>[16][17]</sup> and Catalyst.<sup>[18][19]</sup>

PyTorch provides two high-level features:<sup>[20]</sup>

- Tensor computing (like NumPy) with strong acceleration via graphics processing units (GPU)
- Deep neural networks built on a tape-based automatic differentiation system

## History

In 2001, Torch was written and released under a GPL license. It was a machine-learning library written in C++, supporting methods including neural networks, SVM, hidden Markov models, etc.<sup>[21][22][23]</sup> It was improved to Torch7 in 2012.<sup>[24]</sup> Development on Torch ceased in 2018 and was subsumed by the PyTorch project.<sup>[25]</sup>

Meta (formerly known as Facebook) operates both PyTorch and Convolutional Architecture for Fast Feature Embedding (Caffe2), but models defined by the two frameworks were mutually incompatible. The Open Neural Network Exchange (ONNX) project was created by Meta and Microsoft in September 2017 for converting models between frameworks. Caffe2 was merged into PyTorch at the end of March 2018.<sup>[26]</sup> In September 2022, Meta announced that PyTorch would be governed by the independent PyTorch Foundation, a newly created subsidiary of the Linux Foundation.<sup>[27]</sup>



|                           |                                                                                                                        |
|---------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Original author(s)</b> | Adam Paszke<br>Sam Gross<br>Soumith Chintala<br>Gregory Chanan                                                         |
| <b>Developer(s)</b>       | Meta AI                                                                                                                |
| <b>Initial release</b>    | September 2016 <sup>[1]</sup>                                                                                          |
| <b>Stable release</b>     | 2.7.1 <sup>[2]</sup>  / 4 June 2025 |
| <b>Repository</b>         | <u>github.com/pytorch/pytorch</u><br>( <u>https://github.com/pytorch/pytorch</u> )                                     |
| <b>Written in</b>         | <u>Python</u><br><u>C++</u><br><u>CUDA</u>                                                                             |
| <b>Operating system</b>   | <u>Linux</u><br><u>macOS</u><br><u>Windows</u>                                                                         |
| <b>Platform</b>           | <u>IA-32</u> , <u>x86-64</u> , <u>ARM64</u>                                                                            |
| <b>Available in</b>       | English                                                                                                                |
| <b>Type</b>               | <u>Library for machine learning and deep learning</u>                                                                  |
| <b>License</b>            | <u>BSD-3</u> <sup>[3]</sup>                                                                                            |
| <b>Website</b>            | <u>pytorch.org</u> ( <u>https://pytorch.org/</u> )                                                                     |

PyTorch 2.0 was released on 15 March 2023, introducing [TorchDynamo](#), a Python-level [compiler](#) that makes code run up to 2x faster, along with significant improvements in training and inference performance across major [cloud platforms](#).<sup>[28][29]</sup>

## PyTorch tensors

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PyTorch defines a class called Tensor (`torch.Tensor`) to store and operate on homogeneous multidimensional rectangular arrays of numbers. PyTorch Tensors are similar to [NumPy Arrays](#), but can also be operated on a [CUDA](#)-capable [NVIDIA GPU](#). PyTorch has also been developing support for other GPU platforms, for example, AMD's [ROCm](#)<sup>[30]</sup> and Apple's [Metal Framework](#).<sup>[31]</sup>

PyTorch supports various sub-types of Tensors.<sup>[32]</sup>

Note that the term "tensor" here does not carry the same meaning as tensor in mathematics or physics. The meaning of the word in machine learning is only superficially related to its original meaning as a certain kind of object in [linear algebra](#). Tensors in PyTorch are simply multi-dimensional arrays.

## PyTorch neural networks

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PyTorch defines a module called nn (`torch.nn`) to describe neural networks and to support training. This module offers a comprehensive collection of building blocks for neural networks, including various layers and activation functions, enabling the construction of complex models. Networks are built by inheriting from the `torch.nn` module and defining the sequence of operations in the `forward()` function.

## Example

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The following program shows the low-level functionality of the library with a simple example.

```
1 import torch
2 dtype = torch.float
3 device = torch.device("cpu") # Execute all calculations on the CPU
4 # device = torch.device("cuda:0") # Executes all calculations on the GPU
5
6 # Create a tensor and fill it with random numbers
7 a = torch.randn(2, 3, device=device, dtype=dtype)
8 print(a)
9 # Output: tensor([[ -1.1884,  0.8498, -1.7129],
10 #                [-0.8816,  0.1944,  0.5847]])
11
12 b = torch.randn(2, 3, device=device, dtype=dtype)
13 print(b)
14 # Output: tensor([[ 0.7178, -0.8453, -1.3403],
15 #                [ 1.3262,  1.1512, -1.7070]])
16
17 print(a * b)
18 # Output: tensor([[ -0.8530, -0.7183,  2.58],
19 #                [-1.1692,  0.2238, -0.9981]])
20
21 print(a.sum())
22 # Output: tensor(-2.1540)
23
24 print(a[1, 2]) # Output of the element in the third column of the second row (zero-based)
```

```
25 # Output: tensor(0.5847)
26
27 print(a.max())
28 # Output: tensor(0.8498)
```

The following code-block defines a neural network with linear layers using the nn module.

```
1 from torch import nn # Import the nn sub-module from PyTorch
2
3 class NeuralNetwork(nn.Module): # Neural networks are defined as classes
4     def __init__(self): # Layers and variables are defined in the __init__ method
5         super().__init__() # Must be in every network.
6         self.flatten = nn.Flatten() # Construct a flattening layer.
7         self.linear_relu_stack = nn.Sequential( # Construct a stack of layers.
8             nn.Linear(28 * 28, 512), # Linear Layers have an input and output shape
9             nn.ReLU(), # ReLU is one of many activation functions provided by nn
10            nn.Linear(512, 512),
11            nn.ReLU(),
12            nn.Linear(512, 10),
13        )
14
15     def forward(self, x): # This function defines the forward pass.
16         x = self.flatten(x)
17         logits = self.linear_relu_stack(x)
18         return logits
```

## See also



**Free and open-  
source software  
portal**

- [Comparison of deep learning software](#)
- [Differentiable programming](#)
- [DeepSpeed](#)

## References

1. Chintala, Soumith (1 September 2016). "PyTorch Alpha-1 release" (<https://github.com/pytorch/pytorch/releases/tag/v0.1.1>). *GitHub*. Archived (<https://web.archive.org/web/20210829055231/https://github.com/pytorch/pytorch/releases/tag/v0.1.1>) from the original on 29 August 2021. Retrieved 19 August 2020.
2. "Release 2.7.1" (<https://github.com/pytorch/pytorch/releases/tag/v2.7.1>). 4 June 2025. Retrieved 15 June 2025.
3. Claburn, Thomas (12 September 2022). "PyTorch gets lit under The Linux Foundation" ([https://www.theregister.com/2022/09/12/pytorch\\_meta\\_linux\\_foundation/](https://www.theregister.com/2022/09/12/pytorch_meta_linux_foundation/)). *The Register*. Archived ([https://web.archive.org/web/20221018040848/https://www.theregister.com/2022/09/12/pytorch\\_meta\\_linux\\_foundation/](https://web.archive.org/web/20221018040848/https://www.theregister.com/2022/09/12/pytorch_meta_linux_foundation/)) from the original on 18 October 2022. Retrieved 18 October 2022.
4. Yegulalp, Serdar (19 January 2017). "Facebook brings GPU-powered machine learning to Python" (<https://www.infoworld.com/article/3159120/artificial-intelligence/facebook-brings-gpu-powered-machine-learning-to-python.html>). *InfoWorld*. Archived (<https://web.archive.org/web/20180712054543/https://www.infoworld.com/article/3159120/artificial-intelligence/facebook-brings-gpu-powered-machine-learning-to-python.html>) from the original on 12 July 2018. Retrieved 11 December 2017.

5. Loriga, Ben (3 August 2017). "Why AI and machine learning researchers are beginning to embrace PyTorch" (<https://www.oreilly.com/ideas/why-ai-and-machine-learning-researchers-are-beginning-to-embrace-pytorch>). O'Reilly Media. Archived (<https://web.archive.org/web/20190517055218/https://www.oreilly.com/ideas/why-ai-and-machine-learning-researchers-are-beginning-to-embrace-pytorch>) from the original on 17 May 2019. Retrieved 11 December 2017.
6. Ketkar, Nikhil (2017). "Introduction to PyTorch". *Deep Learning with Python*. Apress, Berkeley, CA. pp. 195–208. doi:10.1007/978-1-4842-2766-4\_12 ([https://doi.org/10.1007%2F978-1-4842-2766-4\\_12](https://doi.org/10.1007%2F978-1-4842-2766-4_12)). ISBN 9781484227657.
7. Moez Ali (June 2023). "NLP with PyTorch: A Comprehensive Guide" (<https://www.datacamp.com/tutorial/nlp-with-pytorch-a-comprehensive-guide>). *datacamp.com*. Archived (<https://web.archive.org/web/20240401214813/https://www.datacamp.com/tutorial/nlp-with-pytorch-a-comprehensive-guide>) from the original on 1 April 2024. Retrieved 1 April 2024.
8. Patel, Mo (7 December 2017). "When two trends fuse: PyTorch and recommender systems" (<https://www.oreilly.com/ideas/when-two-trends-fuse-pytorch-and-recommender-systems>). O'Reilly Media. Archived (<https://web.archive.org/web/20190330131436/https://www.oreilly.com/ideas/when-two-trends-fuse-pytorch-and-recommender-systems>) from the original on 30 March 2019. Retrieved 18 December 2017.
9. Mannes, John. "Facebook and Microsoft collaborate to simplify conversions from PyTorch to Caffe2" (<https://techcrunch.com/2017/09/07/facebook-and-microsoft-collaborate-to-simplify-conversions-from-pytorch-to-caffe2/>). *TechCrunch*. Archived (<https://web.archive.org/web/20200706115906/https://techcrunch.com/2017/09/07/facebook-and-microsoft-collaborate-to-simplify-conversions-from-pytorch-to-caffe2/>) from the original on 6 July 2020. Retrieved 18 December 2017. "FAIR is accustomed to working with PyTorch – a deep learning framework optimized for achieving state of the art results in research, regardless of resource constraints. Unfortunately in the real world, most of us are limited by the computational capabilities of our smartphones and computers."
10. Arakelyan, Sophia (29 November 2017). "Tech giants are using open source frameworks to dominate the AI community" (<https://venturebeat.com/2017/11/29/tech-giants-are-using-open-source-frameworks-to-dominate-the-ai-community/>). *VentureBeat*. Archived (<https://web.archive.org/web/20190330131432/https://venturebeat.com/2017/11/29/tech-giants-are-using-open-source-frameworks-to-dominate-the-ai-community/>) from the original on 30 March 2019. Retrieved 18 December 2017.
11. "PyTorch strengthens its governance by joining the Linux Foundation" (<https://pytorch.org/blog/PyTorchfoundation/>). *pytorch.org*. Retrieved 13 September 2022.
12. "Top 30 Open Source Projects" (<https://github.com/cncf/velocity>). *Open Source Project Velocity by CNCF*. Archived (<https://web.archive.org/web/20230903024925/https://github.com/cncf/velocity>) from the original on 3 September 2023. Retrieved 12 October 2023.
13. "The C++ Frontend" (<https://pytorch.org/cppdocs/frontend.html>). *PyTorch Master Documentation*. Archived (<https://web.archive.org/web/20190729202037/https://pytorch.org/cppdocs/frontend.html>) from the original on 29 July 2019. Retrieved 29 July 2019.
14. Karpathy, Andrej (6 November 2019). "PyTorch at Tesla - Andrej Karpathy, Tesla" (<https://www.youtube.com/watch?v=oBklltKXtDE>). *YouTube*. Archived (<https://web.archive.org/web/20230324144838/https://www.youtube.com/watch?v=oBklltKXtDE>) from the original on 24 March 2023. Retrieved 2 June 2020.
15. "Uber AI Labs Open Sources Pyro, a Deep Probabilistic Programming Language" (<https://eng.uber.com/pyro/>). *Uber Engineering Blog*. 3 November 2017. Archived (<https://web.archive.org/web/20171225034106/https://eng.uber.com/pyro/>) from the original on 25 December 2017. Retrieved 18 December 2017.
16. *PYTORCH-TRANSFORMERS: PyTorch implementations of popular NLP Transformers* ([https://pytorch.org/hub/huggingface\\_pytorch-transformers/](https://pytorch.org/hub/huggingface_pytorch-transformers/)), PyTorch Hub, 1 December 2019, archived ([https://web.archive.org/web/20230611061047/https://pytorch.org/hub/huggingface\\_pytorch-transformers/](https://web.archive.org/web/20230611061047/https://pytorch.org/hub/huggingface_pytorch-transformers/)) from the original on 11 June 2023, retrieved 1 December 2019

17. "Ecosystem Tools" (<https://pytorch.org/ecosystem/>). *pytorch.org*. Archived (<https://web.archive.org/web/20230718105354/https://pytorch.org/ecosystem/>) from the original on 18 July 2023. Retrieved 18 June 2020.
18. *GitHub - catalyst-team/catalyst: Accelerated DL & RL* (<https://github.com/catalyst-team/catalyst>), Catalyst-Team, 5 December 2019, archived (<https://web.archive.org/web/20191222162045/https://github.com/catalyst-team/catalyst>) from the original on 22 December 2019, retrieved 5 December 2019
19. "Ecosystem Tools" (<https://pytorch.org/ecosystem/>). *pytorch.org*. Archived (<https://web.archive.org/web/20230718105354/https://pytorch.org/ecosystem/>) from the original on 18 July 2023. Retrieved 4 April 2020.
20. "PyTorch – About" (<https://web.archive.org/web/20180615190804/https://pytorch.org/about/>). *pytorch.org*. Archived from the original (<https://pytorch.org/about/>) on 15 June 2018. Retrieved 11 June 2018.
21. "Torch Tutorial", Ronan Collobert, IDIAP, 2002-10-02 (<http://torch.ch/torch3/matos/tutorial.pdf>)
22. R. Collobert, S. Bengio and J. Mariéthoz. Torch: a modular machine learning software library (<https://infoscience.epfl.ch/server/api/core/bitstreams/7513f344-91b6-427d-a020-7836b150a150/content>). Technical Report IDIAP-RR 02-46, IDIAP, 2002.
23. <https://web.archive.org/web/20011031104036/http://www.torch.ch/>
24. Collobert, Ronan; Kavukcuoglu, Koray; Farabet, Clément (2012), Montavon, Grégoire; Orr, Geneviève B.; Müller, Klaus-Robert (eds.), "Implementing Neural Networks Efficiently" ([https://doi.org/10.1007/978-3-642-35289-8\\_28](https://doi.org/10.1007/978-3-642-35289-8_28)), *Neural Networks: Tricks of the Trade: Second Edition*, Berlin, Heidelberg: Springer, pp. 537–557, doi:10.1007/978-3-642-35289-8\_28 ([https://doi.org/10.1007%2F978-3-642-35289-8\\_28](https://doi.org/10.1007%2F978-3-642-35289-8_28)), ISBN 978-3-642-35289-8, retrieved 10 June 2025
25. torch/torch7, Commit fd0ee3b, 2018-07-02 (<https://github.com/torch/torch7/commit/fd0ee3bbf7bfdd21ab10c5ee70b74afaef9409e1>)
26. "Caffe2 Merges With PyTorch" (<https://medium.com/@Synced/caffe2-merges-with-pytorch-a89c70ad9eb7>). 2 April 2018. Archived (<https://web.archive.org/web/20190330143500/https://medium.com/@Synced/caffe2-merges-with-pytorch-a89c70ad9eb7>) from the original on 30 March 2019. Retrieved 2 January 2019.
27. Edwards, Benj (12 September 2022). "Meta spins off PyTorch Foundation to make AI framework vendor neutral" (<https://arstechnica.com/information-technology/2022/09/meta-spins-off-pytorch-foundation-to-make-ai-framework-vendor-neutral/>). *Ars Technica*. Archived (<https://web.archive.org/web/20220913034850/https://arstechnica.com/information-technology/2022/09/meta-spins-off-pytorch-foundation-to-make-ai-framework-vendor-neutral/>) from the original on 13 September 2022. Retrieved 13 September 2022.
28. "Dynamo Overview" ([https://pytorch.org/docs/stable/torch.compiler\\_dynamo\\_overview.html](https://pytorch.org/docs/stable/torch.compiler_dynamo_overview.html)).
29. "PyTorch 2.0 brings new fire to open-source machine learning" (<https://venturebeat.com/ai/pytorch-2-0-brings-new-fire-to-open-source-machine-learning/>). *VentureBeat*. 15 March 2023. Archived (<https://web.archive.org/web/20230316004808/https://venturebeat.com/ai/pytorch-2-0-brings-new-fire-to-open-source-machine-learning/>) from the original on 16 March 2023. Retrieved 16 March 2023.
30. "Installing PyTorch for ROCm" (<https://rocm.docs.amd.com/projects/install-on-linux/en/latest/how-to/3rd-party/pytorch-install.html>). *rocm.docs.amd.com*. 9 February 2024.
31. "Introducing Accelerated PyTorch Training on Mac" (<https://pytorch.org/blog/introducing-accelerated-pytorch-training-on-mac/>). *pytorch.org*. Archived (<https://web.archive.org/web/20240129141050/https://pytorch.org/blog/introducing-accelerated-pytorch-training-on-mac/>) from the original on 29 January 2024. Retrieved 4 June 2022.

32. "An Introduction to PyTorch – A Simple yet Powerful Deep Learning Library" (<https://www.analyticsvidhya.com/blog/2018/02/pytorch-tutorial/>). *analyticsvidhya.com*. 22 February 2018. Archived (<https://web.archive.org/web/20191022200858/https://www.analyticsvidhya.com/blog/2018/02/pytorch-tutorial/>) from the original on 22 October 2019. Retrieved 11 June 2018.

## External links

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- [Official website \(https://pytorch.org\)](https://pytorch.org)
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